

The background of the slide is a collage of several smartphone screens displaying the HYPE app interface. The screens are tilted at various angles, creating a dynamic, layered effect. The app's interface includes a top status bar with the time and battery level, a main header with the HYPE logo, and several sections for account management, transactions, and card services. The text 'TOPIC DETECTION in App Reviews' is prominently displayed in the center, with 'TOPIC DETECTION' in red and 'in App Reviews' in black. Below this, the subtitle 'A Data-Driven Solution for HYPE' is written in a smaller, black font.

TOPIC DETECTION in App Reviews

A Data-Driven Solution for HYPE



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OBJECTIVES of Our Solution:

KEY FEATURES

Automatically analyzes every HYPE app review with EDA to discover patterns

Detects **negative** topics (bugs, complaints, usability issues)

Identifies **recurring problems** and emerging trends

Tags each review **consistently** and at scale

BENEFITS

Faster insights for the Research & Customer Insight team

Reduced manual reading and tagging **effort**

Summarizes large volumes of feedback in seconds

Improved visibility on customer **pain points**

AI TOOL FOR FEEDBACK AUTOMATION



Defining the **VALUE PROPOSITION**

8 DECENT WORK AND ECONOMIC GROWTH



By using automated NLP, we enhance feedback analysis **efficiency**, reduce **manual effort**, and **accelerate** decision-making. Real-time insights from user reviews enable data-driven product improvements, **driving growth** and fostering a **more sustainable** business environment.

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE



By utilizing cutting-edge **AI technology**, we automate topic detection, improving **operational efficiency** and modernizing product development processes. This supports the growth of a more **data-driven** and **technologically advanced infrastructure**.



From Tasks to **DEVELOPMENT CHOICES**



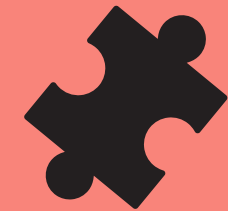
1. EDA & Pre-processing

Exploratory data analysis and preprocessing of text and features.



2. Topic Detection

Detecting topics and comparing them with the HYPE taxonomy.

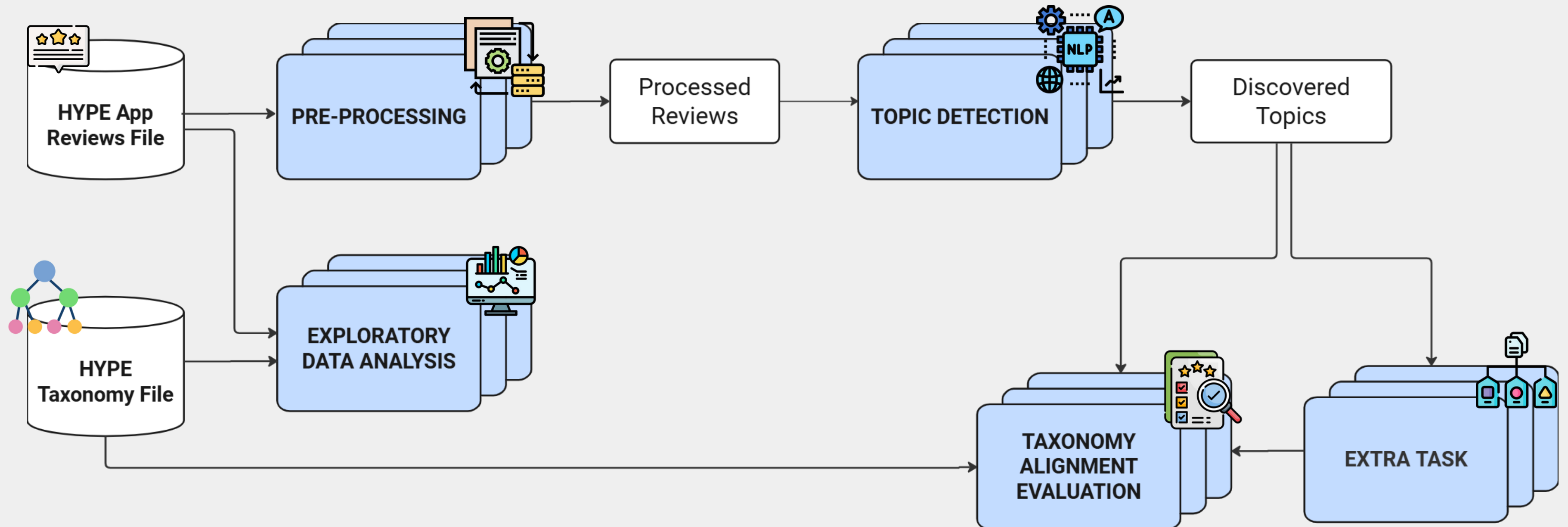


3. Extra Tasks

Future work: multi-label classification and hierarchical topic detection.

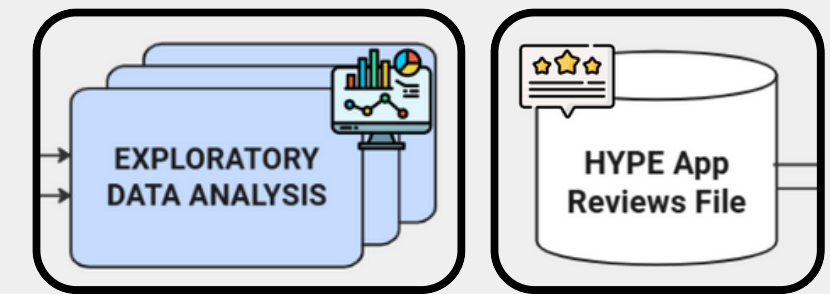
Functional Diagram

SYSTEM OVERVIEW



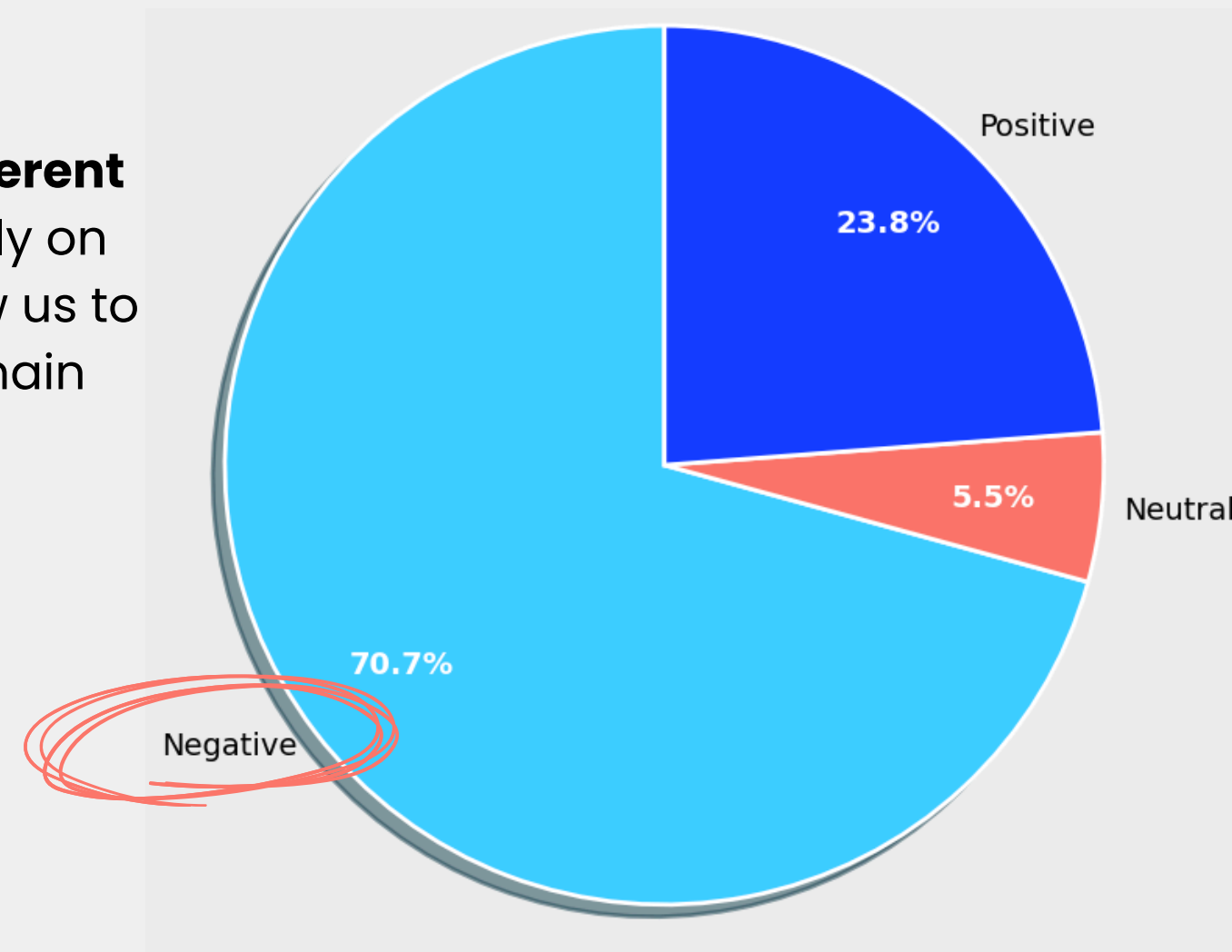
METHOD...

Introducing the DATASET



id_review	review	sentiment	rating	labels	is_labels_	model_version	review_date
00000038-3b3b-5102-be9a-7cf800000000	Nell'applicazione (ricarica conto) non ho più la possibilità di ricaricare in contanti non esiste più l'opzione è cambiato qualcosa?	Negative	5	["problemi_ricarica"]	FALSO	1.0.0	01/10/2024

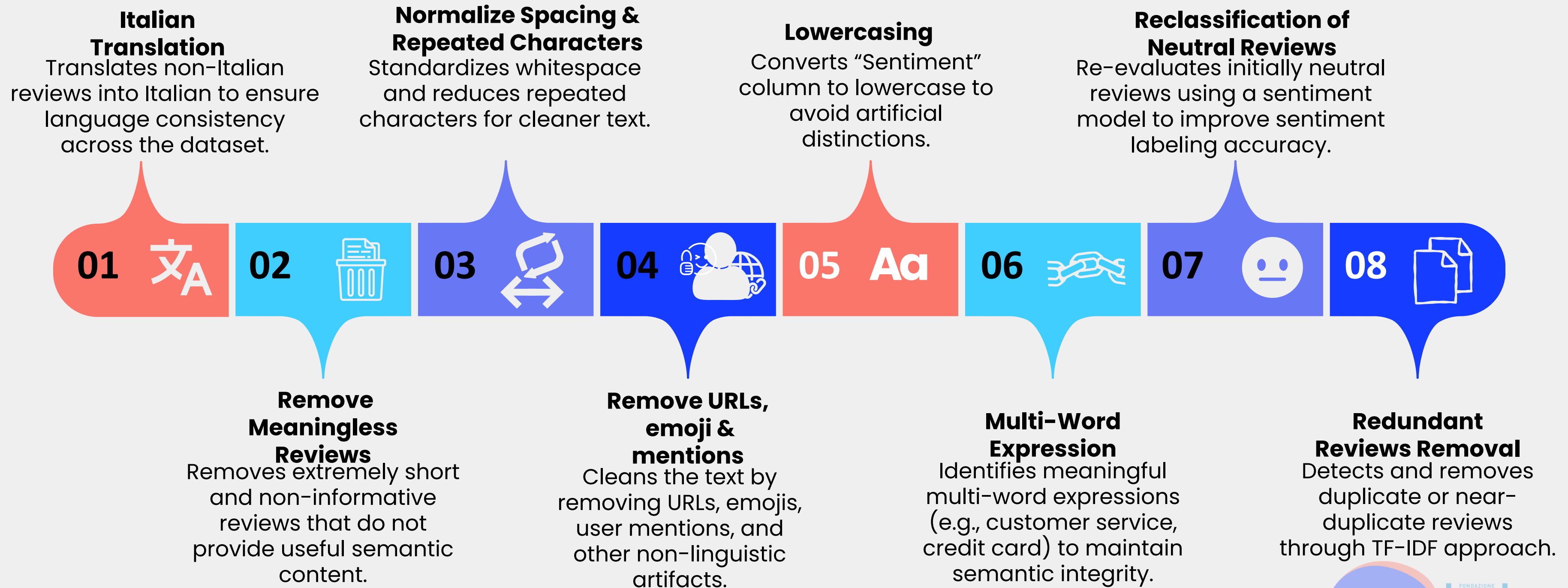
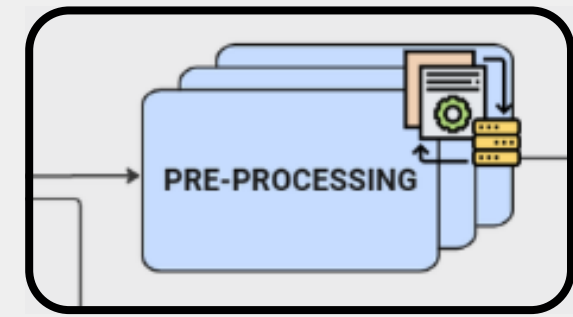
Our data resource consists of **6,150 records**, covering, nowadays, **20 different labels**. We need to focus specifically on the **negative reviews**, as they allow us to identify critical feedback and the main areas that require improvement



From the provided initial dataset, we performed a **series of transformations** to refine it.



PRE-PROCESSING Workflow

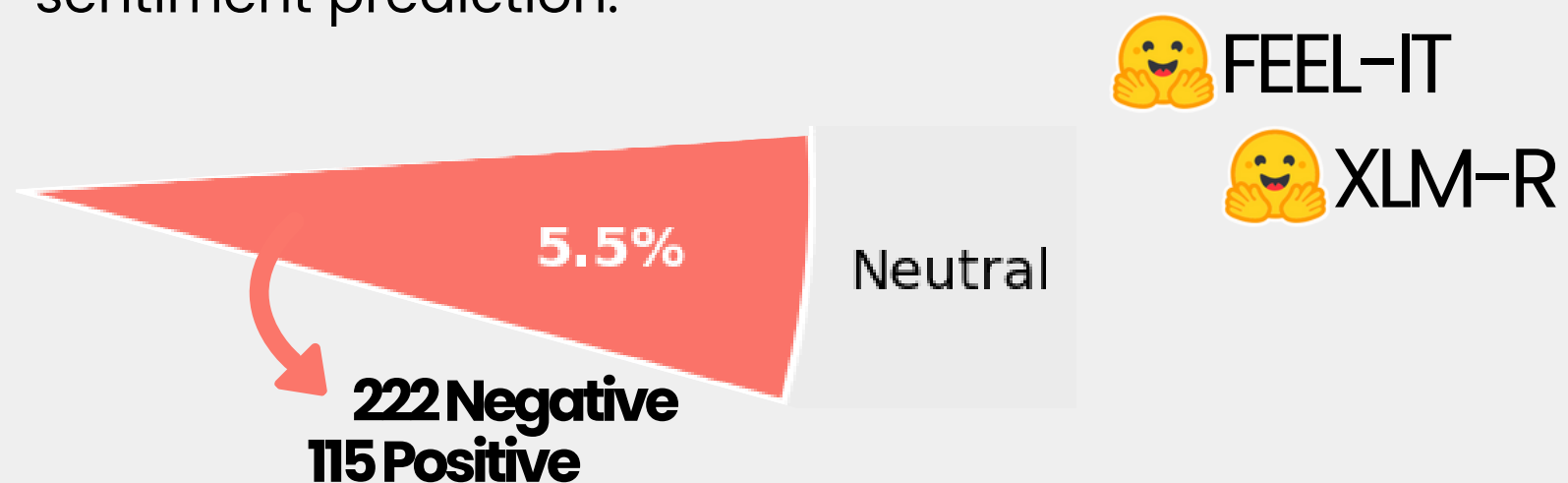


07

Apro l'app più volte a settimana e ogni volta mi chiede di inserire la password e poi parte il Face ID.	
Vi prego fate qualcosa	Neutral
sì deve sempre migliorare	Neutral

Neutral ≠ always problem-free

The initial neutral label does not always reflect the true nature of user feedback. This inconsistency motivates our dataset refinement. We therefore performed a **sentiment-analysis step** using two models and aggregated their outputs to obtain a more reliable sentiment prediction.



 FEEL-IT
 XLM-R

08

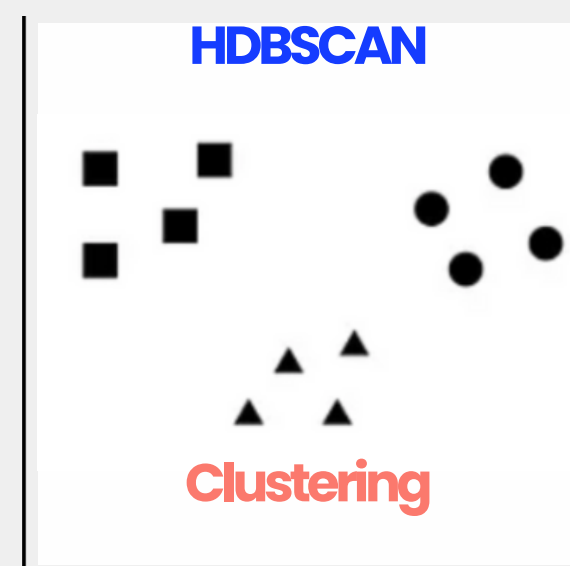
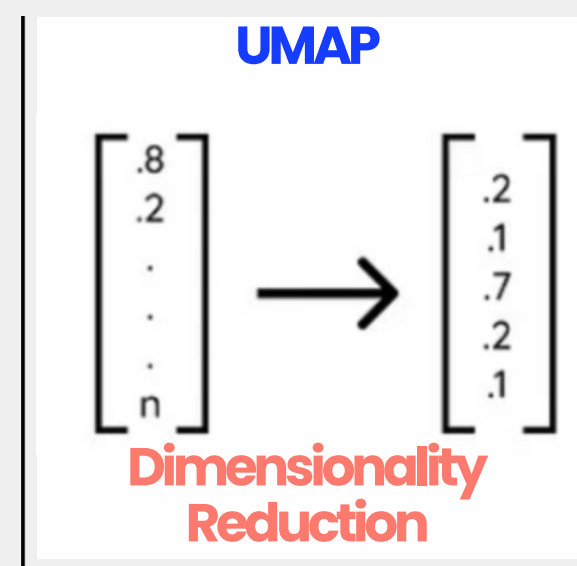
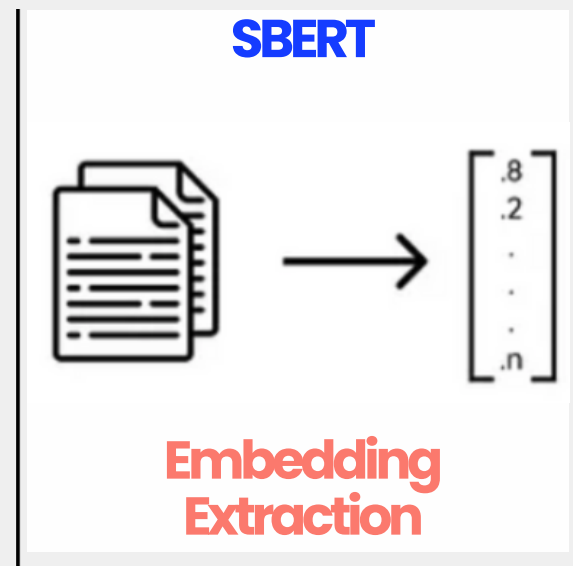
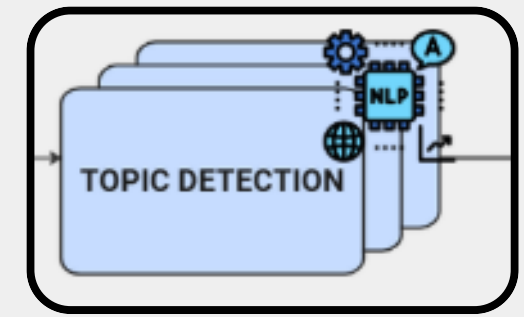


Review 2: Non proteggono dalle frodi, dicendo di bloccare la carta e di emetterne una nuova e a pagamento

Review 1: Chiuderò il conto Hype. Non proteggono dalle frodi, dicendo di bloccare la carta e di emetterne una nuova e a pagamento

After comparing TF-IDF, SBERT, and E5, we opted for a basic **TF-IDF approach**: while the embedding models performed well semantically, they reduced the dataset too aggressively, making TF-IDF the most suitable choice for preserving coverage. This analysis reduce our dataset to **4454 negative** reviews.

Topic Detection BERTopic



CountVectorizer

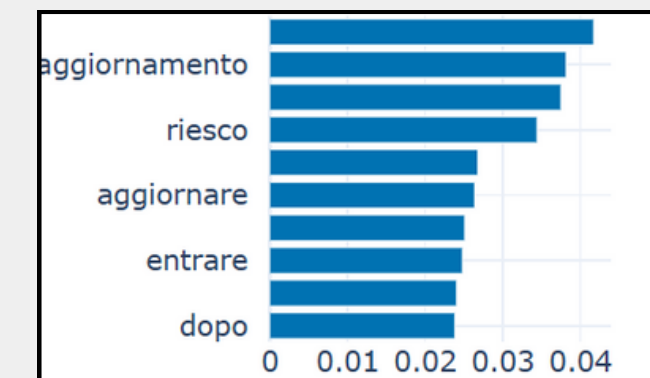
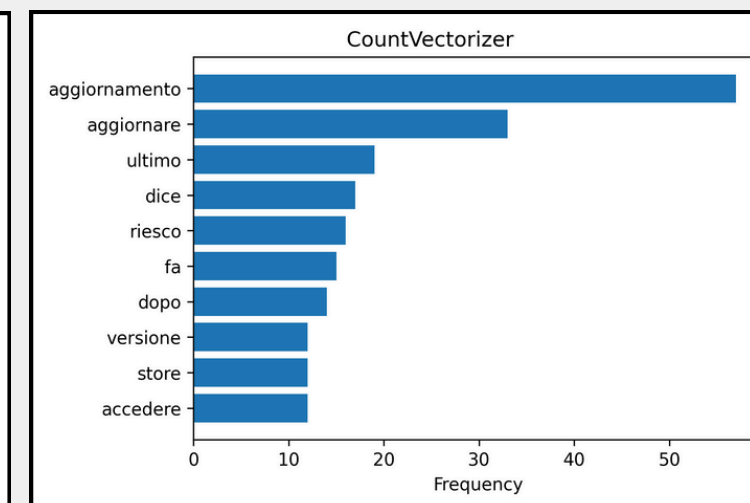
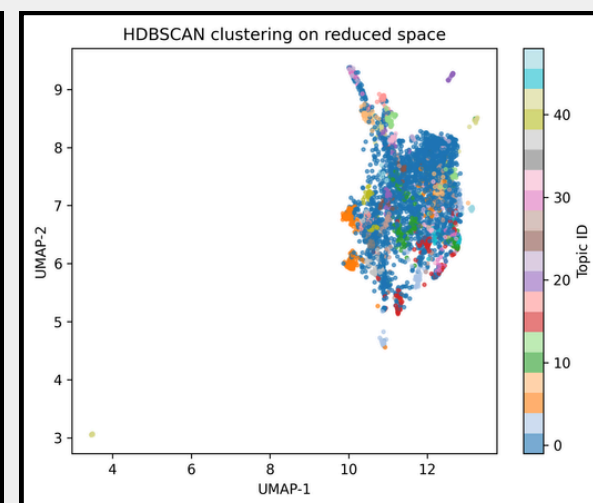
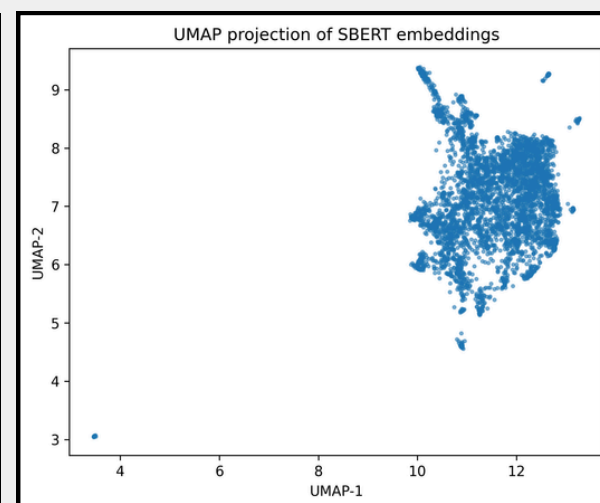
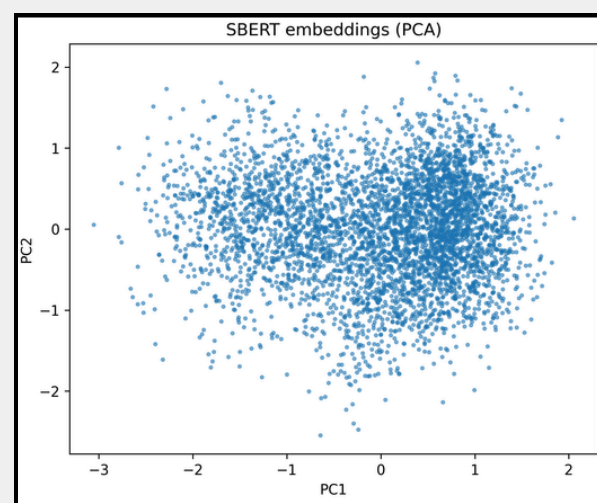
	Token ₁	...	Token _n
Cluster ₁	12	...	80
...	...	...	...
Cluster _m	54	...	37

Bag Of words

c-TF-IDF

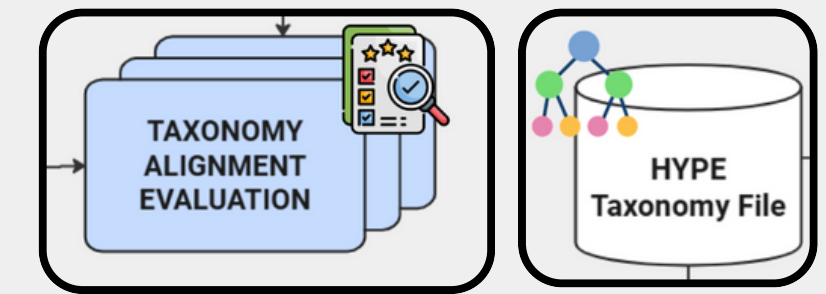
$$\| \text{tf}_{x,c} \|_1 \times \log \left(\frac{A}{f_x} \right)$$

Topic Representation



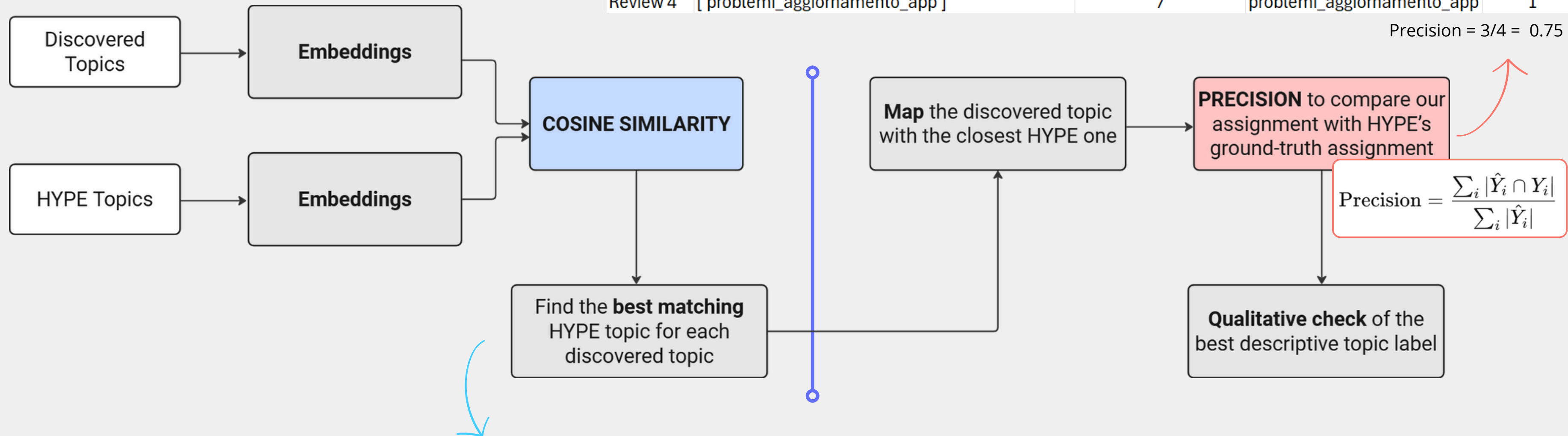
“App Update Issues”

Aligning Topics to the HYPETAXONOMY



EVALUATION AND COMPARISON

TAXONOMY ALIGNMENT

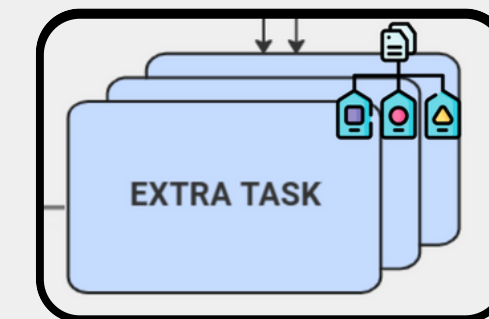


REVIEW	HYPE_TOPIC	DISCOVERED_TOPIC	BEST_MATCH_TOPIC	#CORRECT
Review 1	['problemi_accesso']	2	problemi_accesso	1
Review 2	['problemi_comunicazione', 'problemi_ricarica']	2	problemi_accesso	0
Review 3	['problemi_comunicazione', 'problemi_ricarica']	3	problemi_comunicazione	1
Review 4	['problemi_aggiornamento_app']	7	problemi_aggiornamento_app	1

DISCOVERED_TOPIC	DISCOVERED_TOPIC_WORDS	BEST_MATCH_TOPIC
2	password accedere cambiare riesco accesso sms entrare errore codice fa	problemi_accesso
-1	outlier	[]
3	chat operatore assistenza parlare bot servizio_clienti mail problema rispondono via	problemi_comunicazione
7	aggiornamento dopo ultimo aggiorna aggiornamenti funziona versione riesco dice continua	problemi_aggiornamento_app

EXTRA TASKS

Multi-Label Classification



Some reviews mention **more than one** issue simultaneously, making single-label topic assignment insufficient.

BERTopic

For each review, we exploit the **topic probability distribution** produced by BERTopic.



A multi-label assignment is performed by selecting up to **three topics** whose probability is at least **85%** of the highest topic score.

App Update Issues: ["aggiornamento", "funziona", "errore"]

P= 0.1301

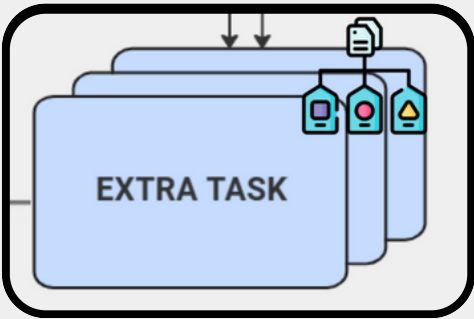
Se potessi non ne metterei nemmeno una più si aggiorna e più da problemi poi contac center praticamente inesistente quando si cercano delle informazioni appena risolto il problema cambierò carta

Support Issues: ["carta", "assistenza", "senza"]

P= 0.1178

EXTRA TASKS

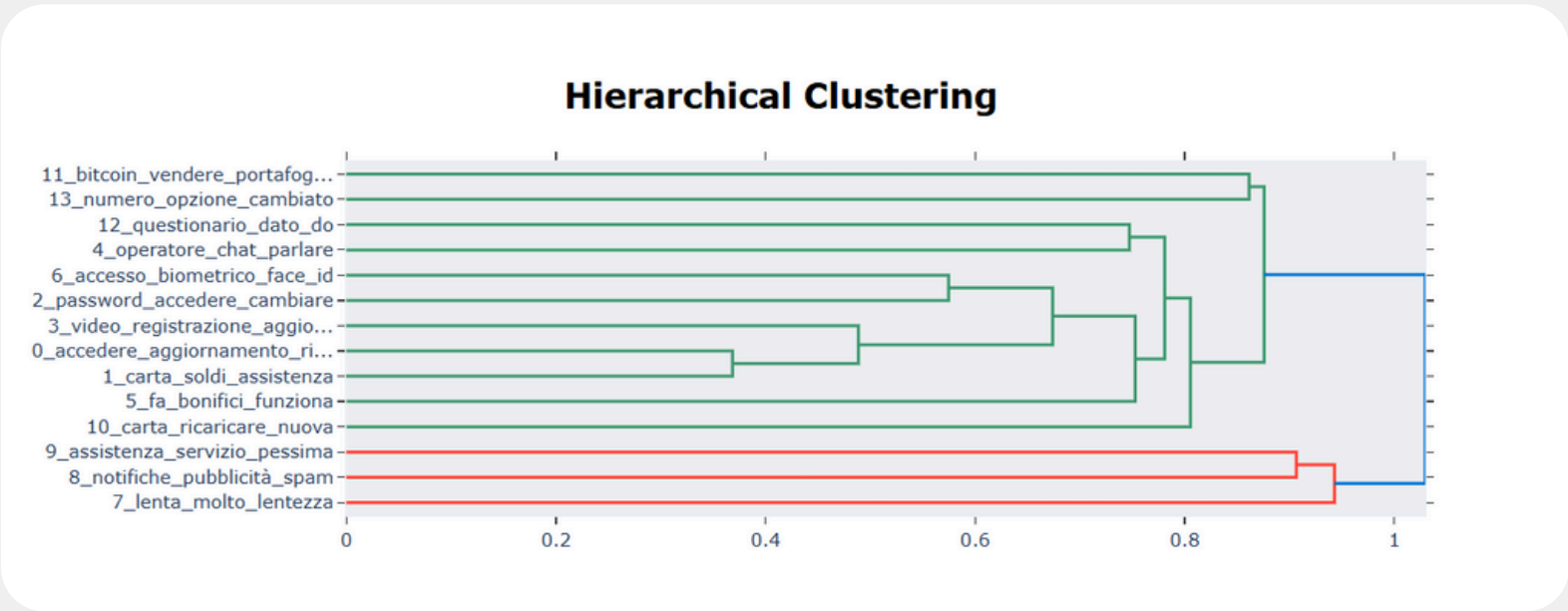
Hierarchical Topic Detection



```
|password_accedere_riescoaccesso_cambiare
|  |■|accesso_biometrico_face_id_valido_riconoscimento — Topic: 6
|  |■|password_accedere_cambiare_riescoaccesso — Topic: 2
|carta_dopo_soldi_assistenza_giorni
|  |■|video_registrazione_aggiornamento_fa_dati — Topic: 3
```



The hierarchical topic structure correctly captures that registration issues, login problems, and post-update access difficulties belong to the same overarching **access-related** problem area.



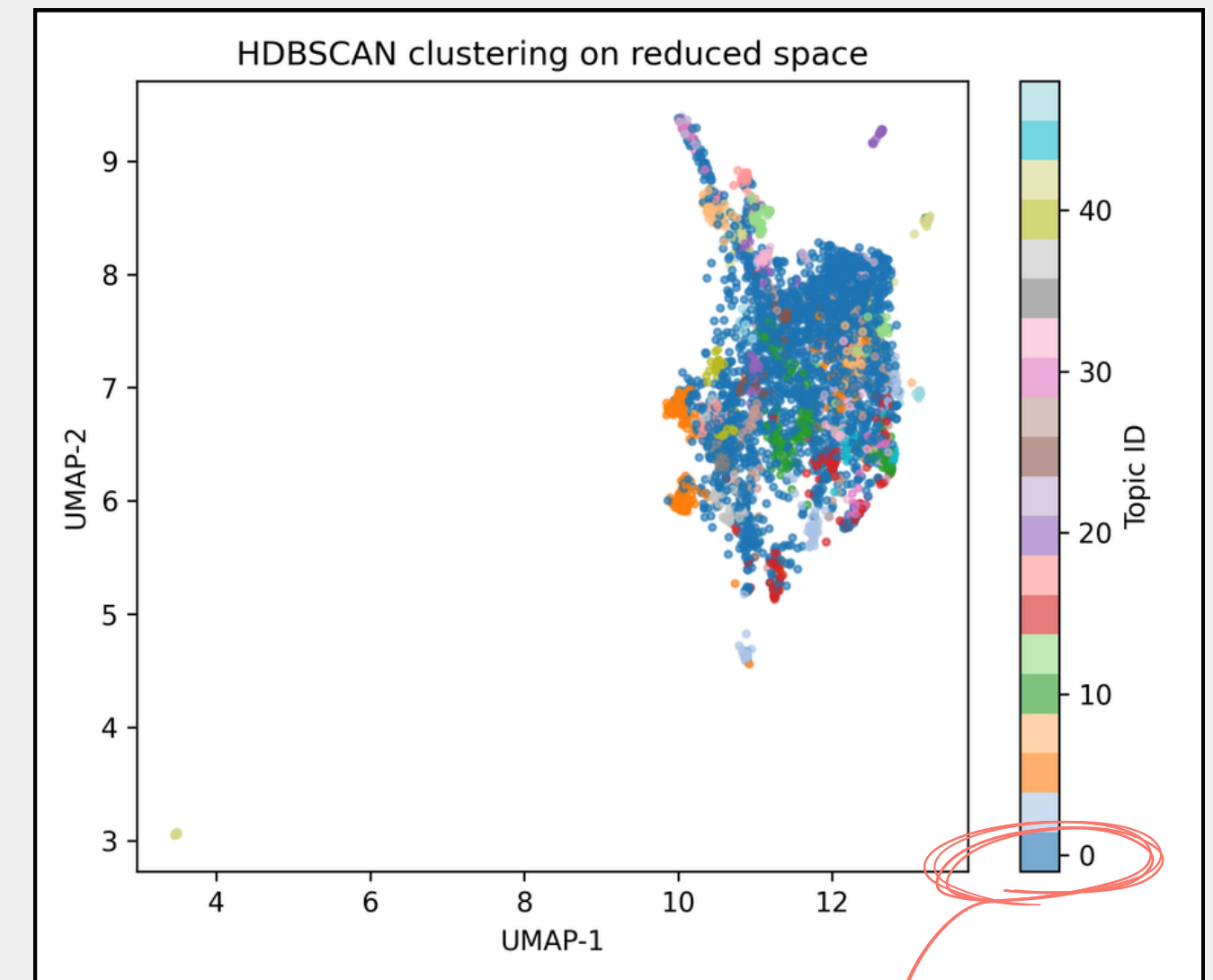
LABEL	PARENT_LABEL
problemi_f24	problemi_pagamenti_generici
problemi_pagamenti_carta	problemi_pagamenti_generici
problemi_bollettini	problemi_pagamenti_generici
problemi_bonifici	problemi_pagamenti_generici

The model shows difficulty in identifying those topics for which the HYPE taxonomy defines an explicit parent label, leading to **limited alignment** at higher hierarchical levels.

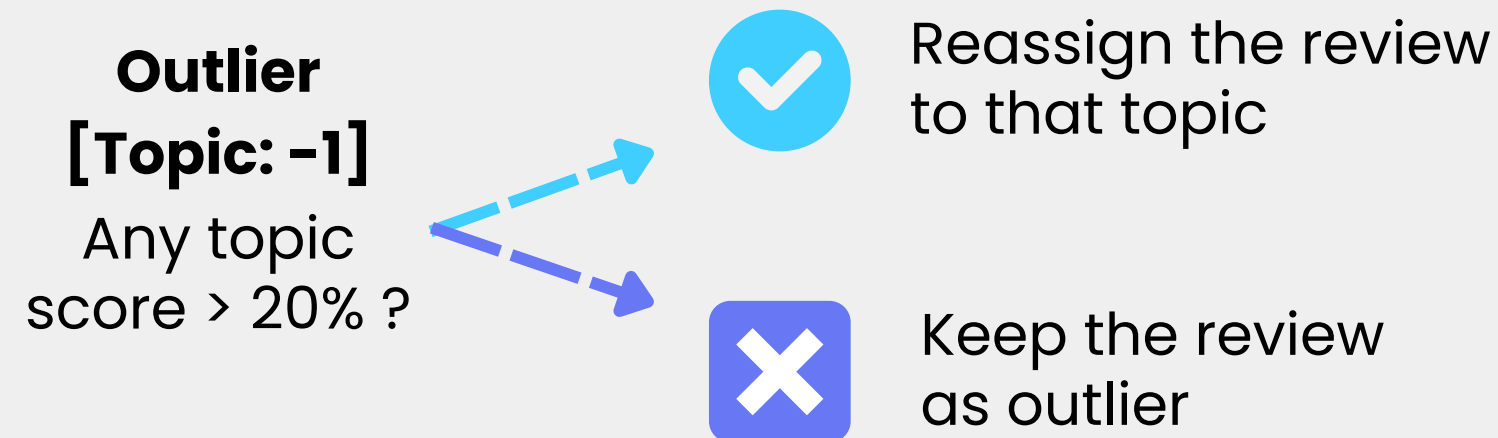
Handling the Excessive NUMBER OF OUTLIERS

Initial topic modeling output produced a high number of outliers.
reducing **coverage** and downstream **interpretability**.

We exploited the multi-label topic analysis computed for each reviews. Outliers are not blindly reassigned, but evaluated using their **secondary topic memberships**.



OUTLIERS



Non riesco più ad aprire l' app .. Crasha !

Labeled as outlier, but

$P(\text{topic} = \text{"Access / App functionality"}) = 0.71$
(keywords: accedere, aggiornamento, riesco, funziona, fa)

HYPOTHESIS AND RESEARCH QUESTIONS

Driving Our Investigation



- **RQ1** — Does machine translation affect the semantic quality of topic detection on multilingual app reviews?
- **RQ2** — How do different stopwords-removal strategies affect the performance of embedding-based topic modeling on app reviews?
- **RQ3** — Which configuration of BERTopic's modular components (embeddings, dimensionality reduction, clustering, and topic representation) yields the best topic quality?
- **RQ4** — Do semi-supervised topic models outperform unsupervised approaches in identifying relevant issues from app reviews?
- **RQ5** — Do embedding-based semantic similarity methods provide reliable mappings between data-driven topics and the existing taxonomy of app issues?

EXPERIMENT & RESULT...

General Evaluation METRICS

Silhouette Score

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

Measures how well documents are clustered by comparing the average **intra-cluster distance** a of each document with the average **inter-cluster distance** b .

Topic Coherence Score

$$\text{Coherence}(t) = \frac{1}{|P|} \sum_{(w_i, w_j) \in P} \cos(\mathbf{e}_{w_i}, \mathbf{e}_{w_j})$$

Measures the semantic consistency of topics by computing, for each topic t , the average **cosine similarity between the embeddings** $\mathbf{e}_{w_i}$ and $\mathbf{e}_{w_j}$ of all pairs of its top representative words.

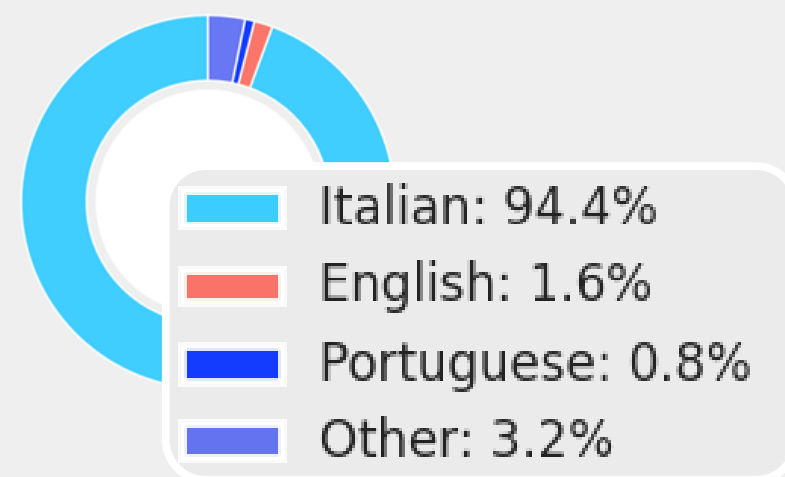
Topic Diversity Score

$$\text{TopicDiversity} = \frac{|W|}{T \cdot K}$$

Measures how different topics are from each other by computing the ratio between the **number of unique words** W and the total number of representative **words across all topics** T , considering the top $K=10$ words per topic.

Handling **MULTILINGUAL REVIEWS**

Method and Results



Some reviews are written in languages **other than Italian**. To ensure consistent analysis, all non-Italian reviews are automatically **translated into Italian** using NLLB.

 **NLLB** – No Language Left Behind

worst app 👎

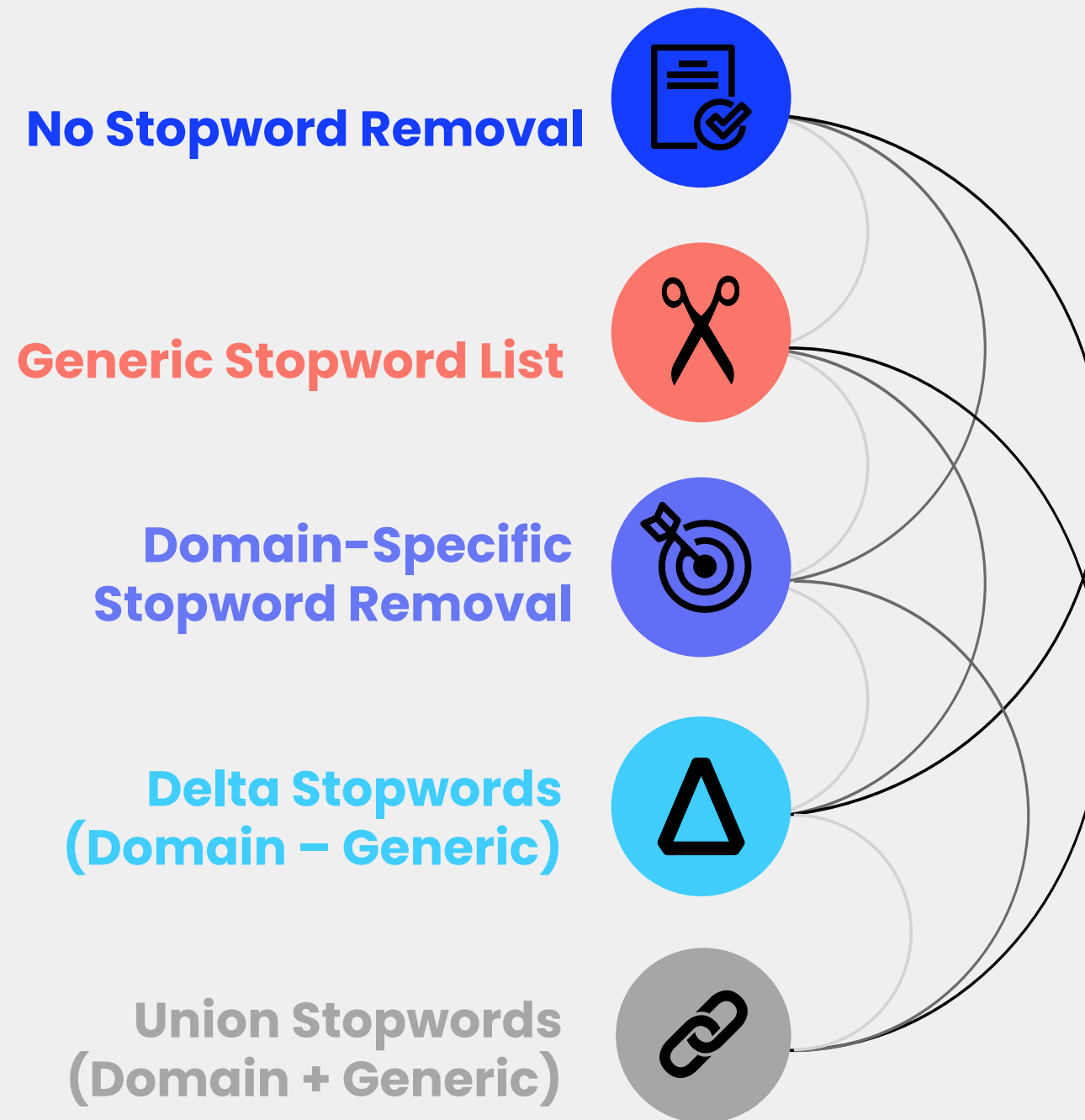


l'app peggiore 👎

Metric	All Reviews	Translated Reviews
Silhouette Score	0.0636	0.068
Precision	0.1773	0.193

➡ **Precision** is computed at document level by comparing the inferred taxonomy labels with the reference labels for each review. It measures the proportion of predicted labels that correctly match the reference label set.

Comparative Evaluation of **STOPWORD STRATEGIES:** Method and Results

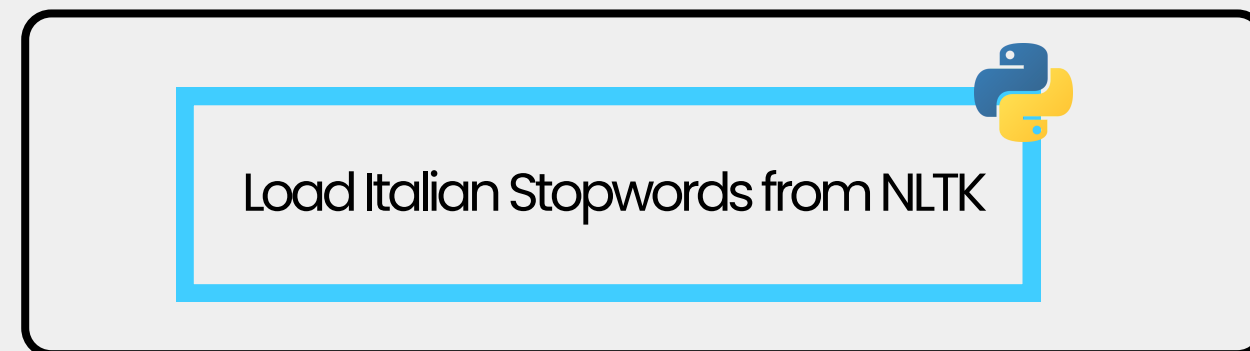
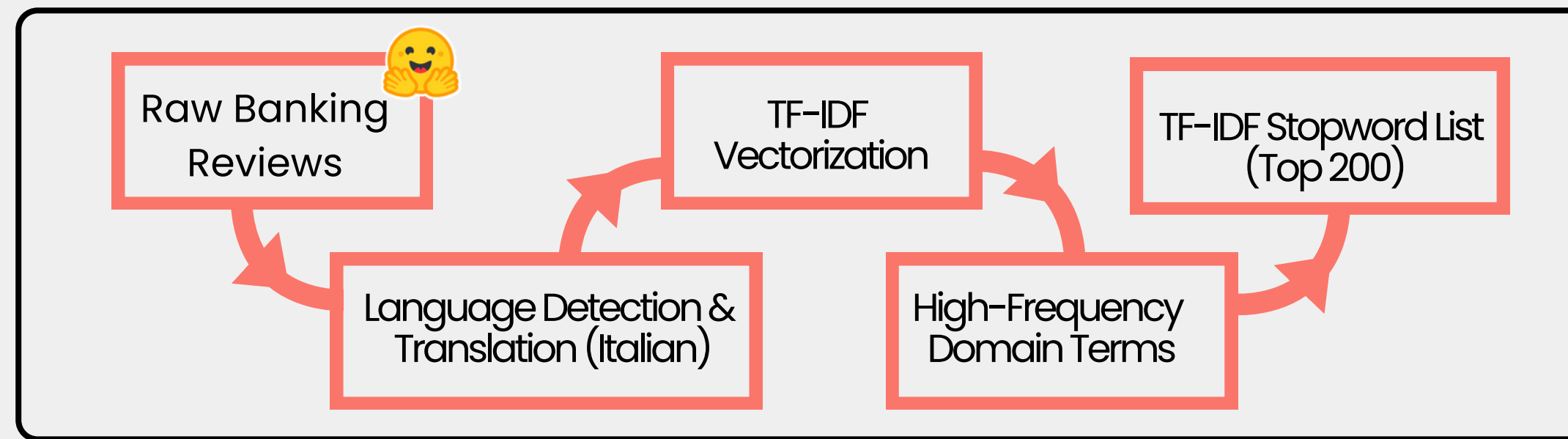


Literature shows that traditional stopwords removal is often **sub-optimal** for embedding-based models, since these models rely on the full syntactic structure of the input text. In our study, we evaluate the impact of different stopwords strategies, including a domain-specific stopwords list generated in a data-driven way using another banking app reviews dataset, to assess whether domain knowledge can **provide benefits**.

[Stop Words for Processing Software Engineering Documents: Do they Matter?]

Comparative Evaluation of **STOPWORD STRATEGIES**: Method and Results

DOMAIN-SPECIFIC STOPWORD LIST



GENERIC STOPWORD LIST



These two stopwords lists are appropriately **combined** and composed to derive the **five stopwords strategies** evaluated in our pipeline.

Comparative Evaluation of **STOPWORD STRATEGIES:** Method and Results

Strategy	Silhouette	Topic Coherence	Topic Diversity
No Stopword Removal	0.0636	0.5375	0.8071
Generic Stopword List	0.0508	0.5313	0.8000
Domain-Specific Stopword Removal	0.0427	0.4993	0.8786
Delta Stopword (Domain - Generic)	0.0389	0.5044	0.8857
Union Stopword	0.0434	0.5236	0.8714

[Results computed using the default approach based on UMAP and DBSCAN.]

Comparing MODELS

LDA

Probabilistic model where the document is a mixture of topics, topic as a distribution over words

PROS: ✓ SIMPLE, FAST, INTERPRETABLE

CONS: ✗ NO SEMANTICS, FIXED #TOPICS,
WEAK ON SHORT TEXT

NMF

Matrix factorization of the document-term matrix into topic components.

PROS: ✓ COHERENT TOPICS, STABLE, GOOD ON
SHORT TEXT

CONS: ✗ NO SEMANTICS, FIXED #TOPICS

BERTopic

Embedding-based model that uses transformer representations, HDBSCAN clustering and c-TF-IDF keyword extraction.

PROS

✓ HIGH **SEMANTIC UNDERSTANDING**

✓ POWERFUL ALSO ON **SHORT AND NOISY TEXT**

✓ DOES NOT REQUIRE STRONG PRE-PROCESSING

CONS

✗ COMPUTATIONALLY HEAVY

✗ SENSITIVE TO EMBEDDING MODEL CHOICE

✗ PRODUCES A SINGLE-TOPIC ASSIGNMENT

Comparing MODELS

LDA

Probabilistic model where the document is a mixture of topics, topic as a distribution over words

Topic Coherence Score: 0.6632

Topic Diversity Score: 0.3333

NMF

Matrix factorization of the document-term matrix into topic components.

Topic Coherence Score: 0.6190

Topic Diversity Score: 0.6867

BERTopic

Embedding-based model that uses transformer representations, HDBSCAN clustering and c-TF-IDF keyword extraction.

Topic Coherence Score: 0.5250

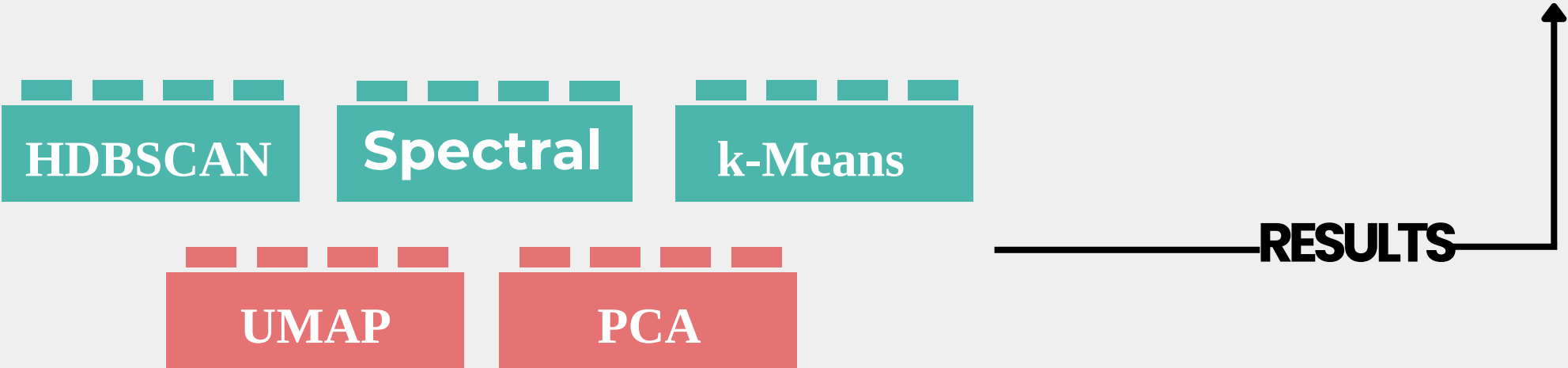
Topic Diversity Score: 0.80714



BERTopic

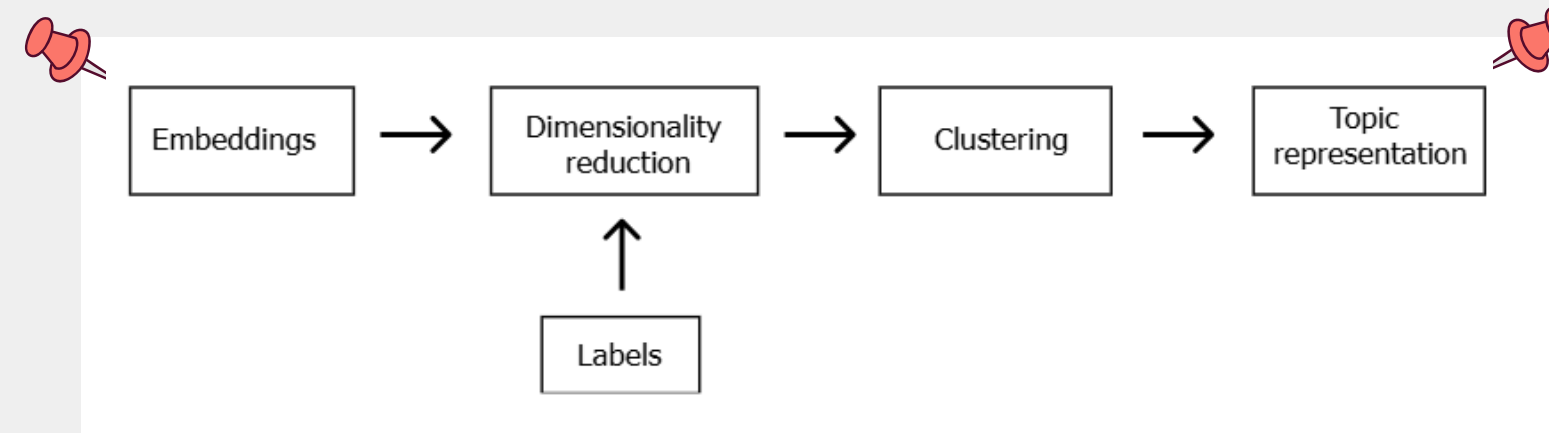


Dim. Reduction	Clustering	Silhouette	Topic Coherence	Topic Diversity
UMAP	HDBSCAN	0.0483	0.5250	0.80714
Kernel PCA	Spectral	0.0241	0.5867	0.4867
Kernel PCA	K-Means	0.0166	0.5698	0.4267
UMAP	Spectral	0.0244	0.5337	0.6933
UMAP	K-Means	0.0187	0.5398	0.6600

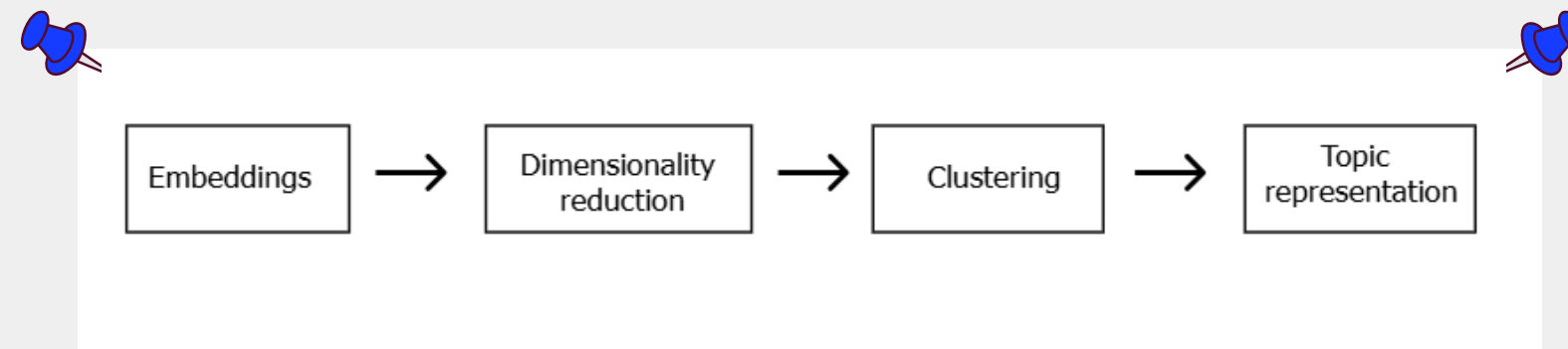


[Results computed using the default approach and hyperparamters, without stopoword removal.]

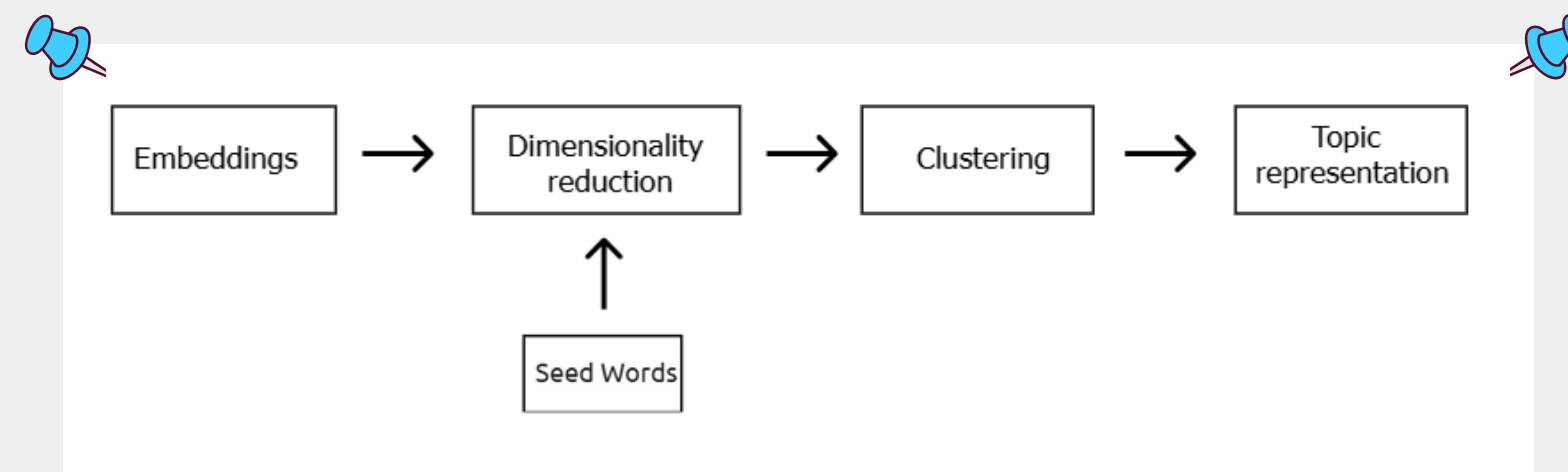
Topic Modeling: THREE APPROACHES COMPARED



SEMI-SUPERVISED TOPIC MODELING

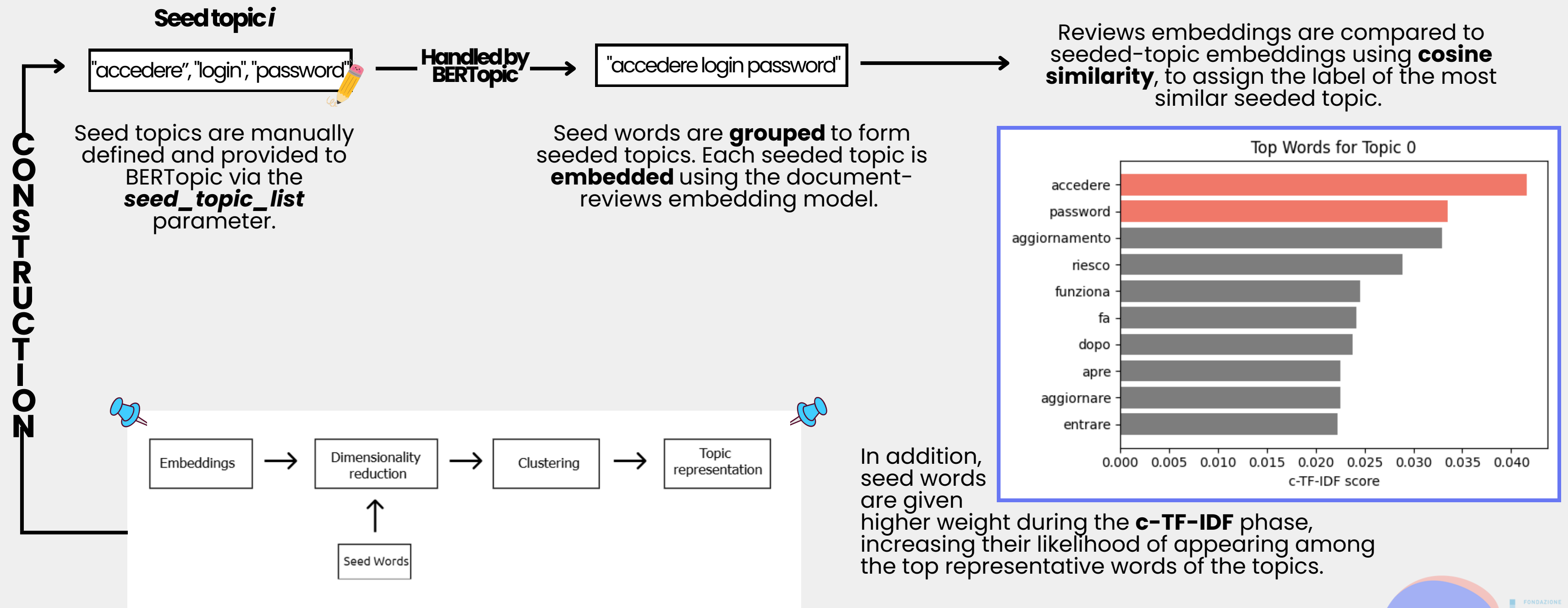


TRADITIONAL TOPIC MODELING



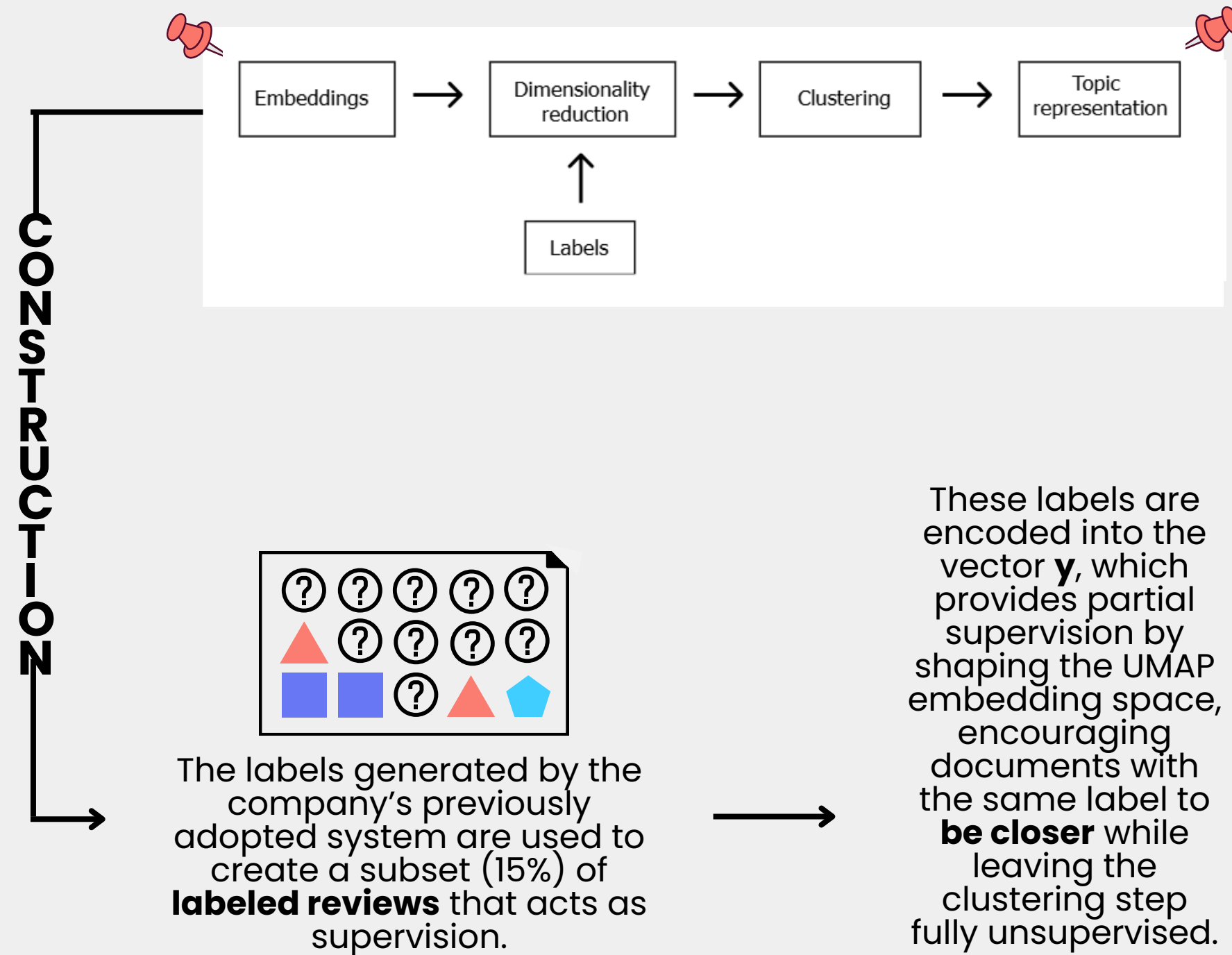
GUIDED TOPIC MODELING

Topic Modeling: THREE APPROACHES COMPARED

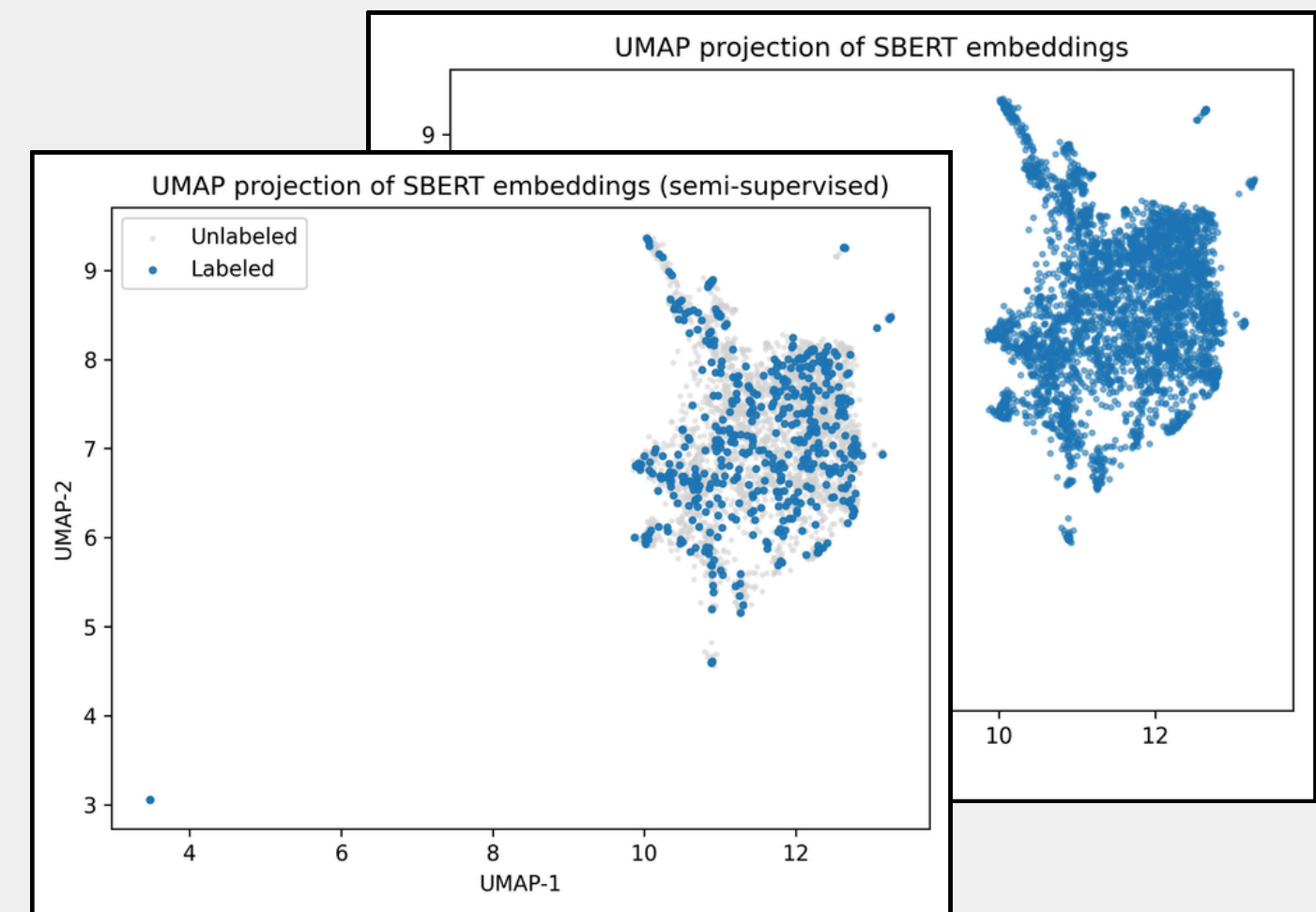


GUIDED TOPIC MODELING

Topic Modeling: THREE APPROACHES COMPARED

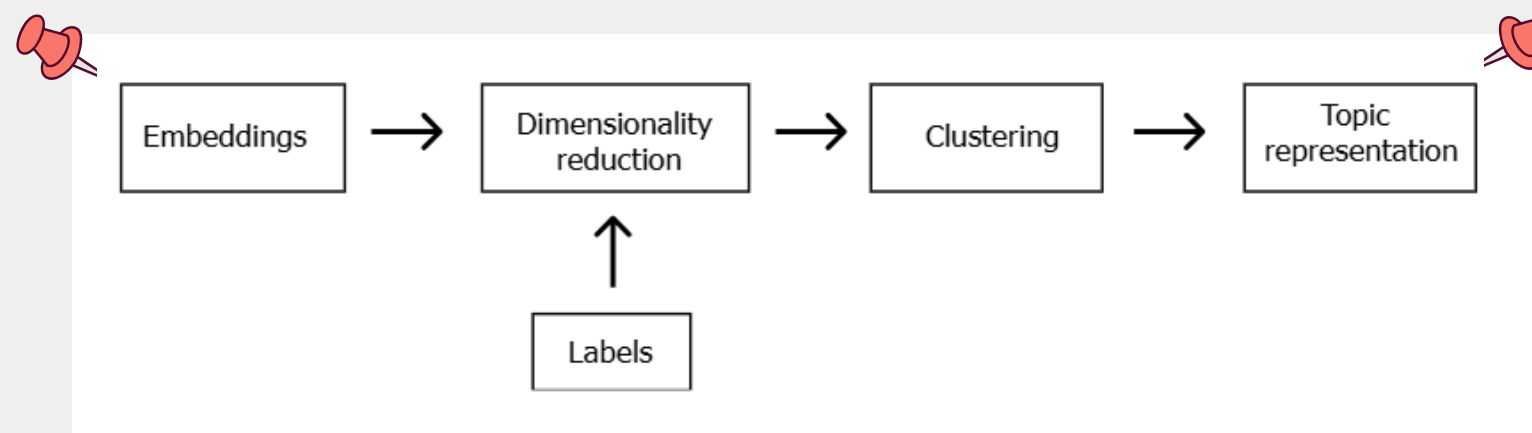


SEMI-SUPERVISED TOPIC MODELING



Topic Modeling: THREE APPROACHES COMPARED

SEMI-SUPERVISED TOPIC MODELING

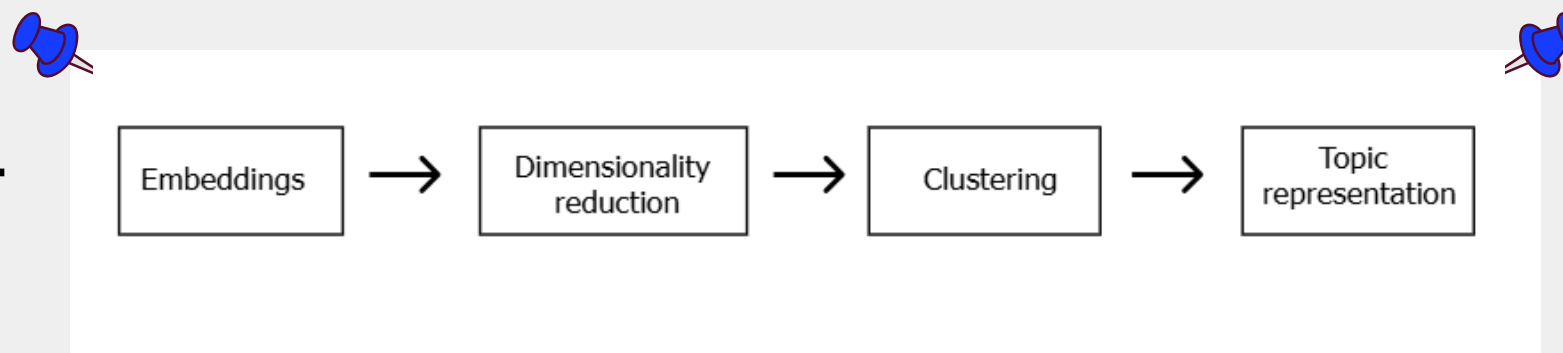
**Silhouette****Topic Coherence****Topic Diversity**

0.0559

0.5336

0.8071

TRADITIONAL TOPIC MODELING

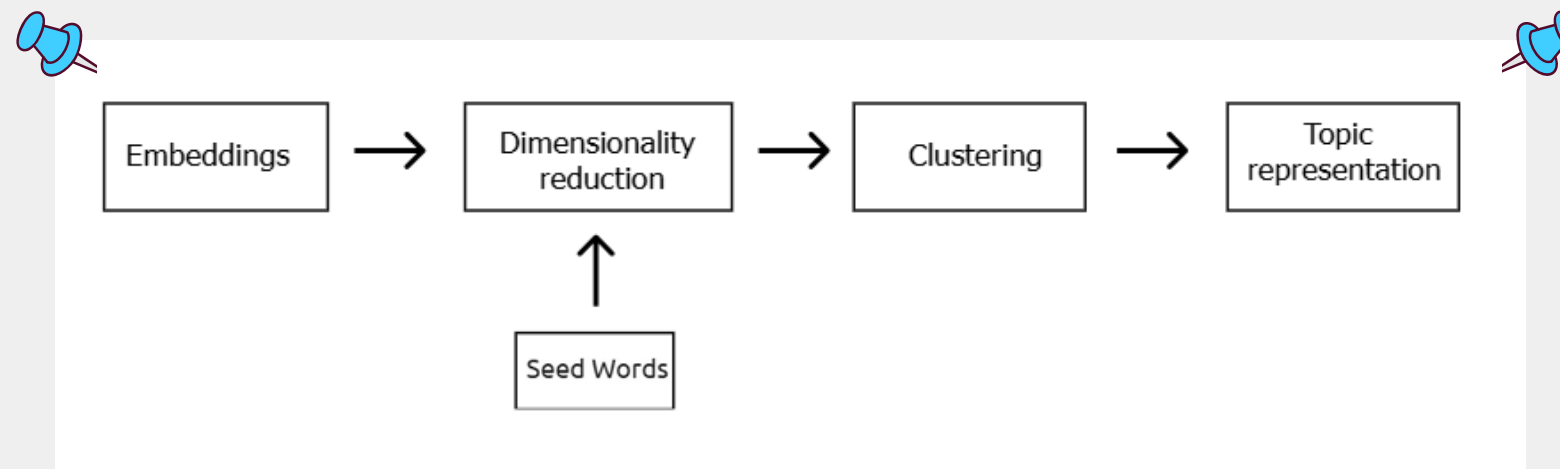


0.0636

0.5375

0.8071

GUIDED TOPIC MODELING



0.0499

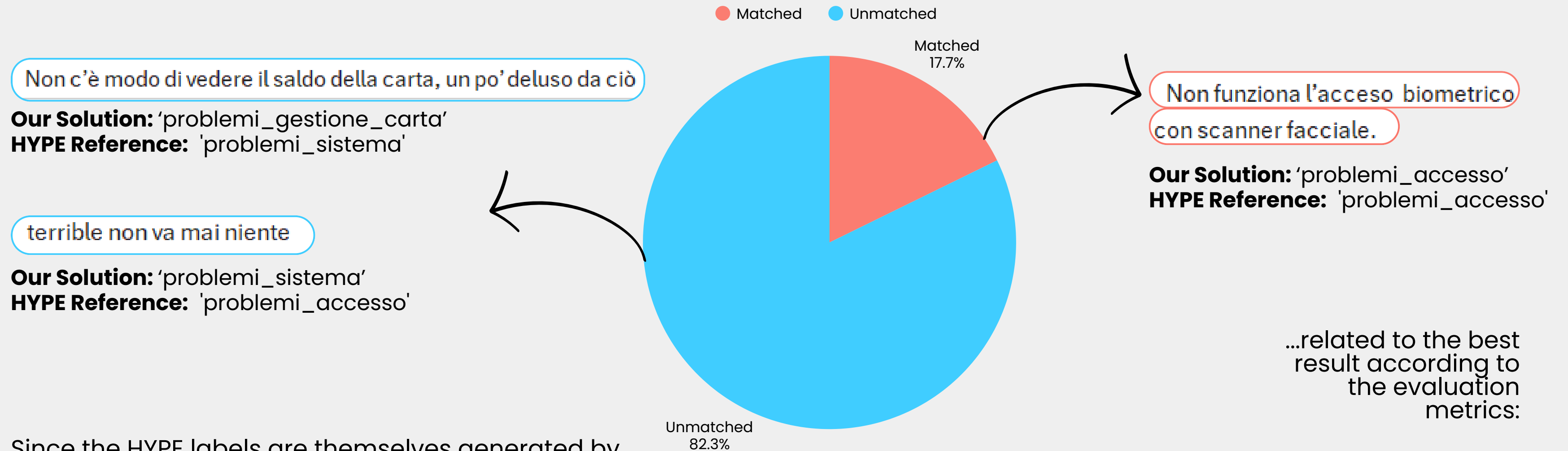
0.4969

0.8071

[Results computed using the default architecture based on UMAP and DBSCAN, without stopwords removal.]

Topic Taxonomy ALIGNMENT RESULTS

Once taxonomy alignment is performed, alignment quality is assessed using a **document-level precision metric**, computed by measuring the overlap between the **inferred taxonomy labels** and the **reference labels** for each review.



Since the HYPE labels are themselves generated by an automated system, a qualitative check allows the identification of cases where the label is marked as **incorrect** by the metric, but the assignment is **accurate**.

...related to the best result according to the evaluation metrics:

BEST CONFIGURATION



Architecture: UMAP + HDBSCAN

Stop-word strategy: None

Topic modeling approach: Traditional

Silhouette	Topic Coherence	Topic Diversity
------------	-----------------	-----------------

0.0636	0.5375	0.8071
--------	--------	--------

UMAP (n_neighbors = 20, n_components = 5, min_dist = 0.0, metric = "cosine")
HDBSCAN (min_cluster_size = 20, min_samples = 5)

By reducing the number of topics, we obtain **14 topics** and we reduce outliers to only 19.12%



0_accedere_aggiornamento_riesco_funziona

1_carta_soldi_assistenza_dopo

2_password_accedere_cambiare_riesco

3_video_registrazione_aggiornamento_fa

4_operatore_chat_parlare_bot

5_fa_bonifici_funziona_mai

6_accesso_biometrico_face_id_valido

7_lenta_molto_lentezza_troppo

8_notifiche_pubblicità_spam_promozioni

9_assistenza_servizio_pessima_assistenza_inesistente

10_carta_ricaricare_nuova_altra

11_bitcoin_vendere_portafoglio_connessione

12_questionario_dato_do_tolgo

13_numero_opzione_cambiato_logicamente

CONCLUSION ...

How we work as a **TEAM**



M
A
N
A
G
E





Testing and comparison of different BERTopic **architectures**, implementation of **hyperparameter tuning** and definition of scoring **metrics**.

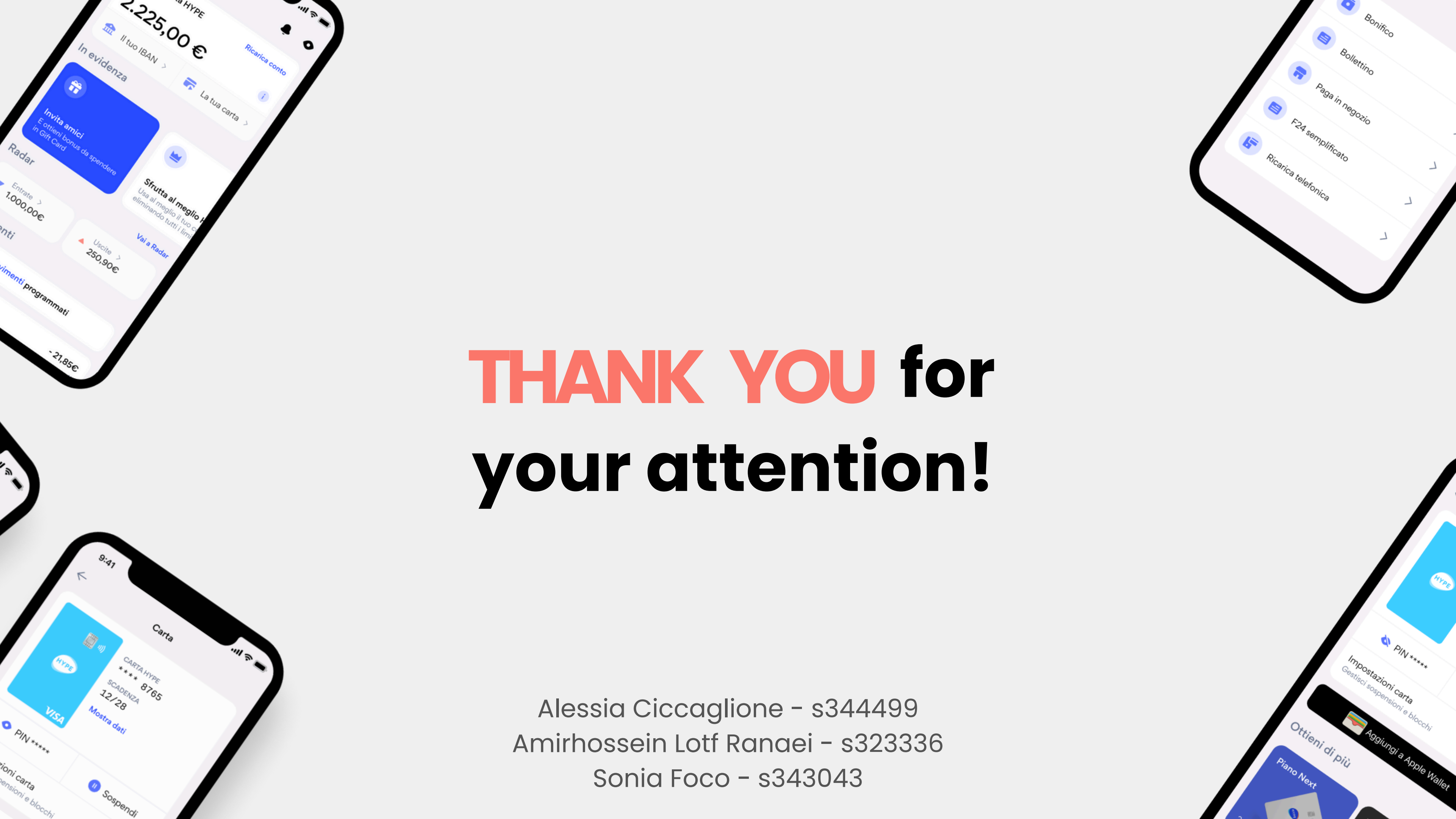


Implementation of advanced **stopword** strategies and different **topic modeling approaches**, development of the **multi-label** topic assignment extra task.



Application and evaluation of **baseline algorithms** as benchmark, development of the **hierarchical** topic detection extra task and organization of the **codebase**

Stay tuned for the
final
PRESENTATION...

The background features a collage of four smartphone screens displaying the HYPE app interface. The top-left screen shows the account balance of 2.225,00 € and various service buttons like 'Invita amici', 'Radar', and 'Sfrutta al meglio'. The top-right screen displays a list of services including 'Bonifico', 'Bollettino', 'Paga in negozio', 'F24 semplificato', and 'Ricarica telefonica'. The bottom-left screen shows a 'Carta' (card) section with a HYPE Visa card and its details. The bottom-right screen shows the 'Impostazioni carta' (card settings) section with options for PIN, card management, and adding to Apple Wallet.

THANK YOU for your attention!

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Amirhossein Lotf Ranaei - s323336
Sonia Foco - s343043